

1. Final answer:

$$\boxed{-1}.$$

Holding the indicated variable constant, the ideal-gas equation gives

$$\left(\frac{\partial V}{\partial T}\right) = \frac{nR}{P}, \quad \left(\frac{\partial T}{\partial P}\right) = \frac{V}{nR}, \quad \left(\frac{\partial P}{\partial V}\right) = -\frac{nRT}{V^2}.$$

Therefore,

$$\frac{nR}{P} \cdot \frac{V}{nR} \cdot \left(-\frac{nRT}{V^2}\right) = -\frac{nRT}{PV} = -1,$$

because $PV = nRT$.

2. Final answer:

$$\boxed{\frac{\partial^2 z}{\partial x^2} = 6yf_u + 36x^2y^2f_{uu} + 48xy^3f_{uv} + 16y^4f_{vv}}.$$

All derivatives of f in this expression are evaluated at

$$(u, v) = (3x^2y, 4xy^2).$$

First,

$$u_x = 6xy, \quad v_x = 4y^2,$$

so the chain rule gives

$$z_x = f_u u_x + f_v v_x = 6xyf_u + 4y^2f_v.$$

Differentiating once more with respect to x ,

$$z_{xx} = f_{uu}u_x^2 + 2f_{uv}u_xv_x + f_{vv}v_x^2 + f_u u_{xx} + f_v v_{xx}.$$

Here

$$u_{xx} = 6y, \quad v_{xx} = 0.$$

Substitution yields

$$z_{xx} = 36x^2y^2f_{uu} + 48xy^3f_{uv} + 16y^4f_{vv} + 6yf_u,$$

which is the displayed answer.

3. **Final answer:**

$$\boxed{\left. \frac{dz}{dt} \right|_{t=1} = 2}.$$

The multivariable chain rule gives

$$\frac{dz}{dt} = f_x(x, y) \frac{dx}{dt} + f_y(x, y) \frac{dy}{dt}.$$

At $t = 1$, $(x, y) = (2, 1)$, $dx/dt = 2t = 2$, and $dy/dt = 2$. Since $\nabla f(2, 1) = (3, -2)$,

$$\left. \frac{dz}{dt} \right|_{t=1} = 3(2) + (-2)(2) = 2.$$

4. **Final answer:**

$$\boxed{\frac{1}{\sqrt{30}}(-5, -1, -2)}.$$

The gradient of the temperature is

$$\nabla T(x, y, z) = (18x + 6yz, -6y + 6xz, 6xy).$$

At the fly's location,

$$\nabla T(2, 2, 2) = (60, 12, 24) = 12(5, 1, 2).$$

The direction of fastest decrease is the negative unit gradient. Since $\|(5, 1, 2)\| = \sqrt{30}$, that direction is

$$-\frac{\nabla T(2, 2, 2)}{\|\nabla T(2, 2, 2)\|} = \frac{1}{\sqrt{30}}(-5, -1, -2).$$